Practical 10

Aim: a) Internetworking with routers in CISCO PACKET TRACER simulator.

d) Design and configure a simple internetwork using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper Straight Through Cable. After forming the network to check network connectivity a simple PDU is transferred from PC0 to PC1

Procedure;

Step -1 Configuring Router 1:

1. Select the router and open CLI.
2. Press ENTER to start configuring Router 1.
3. Type enable to activate the Privileged mode.

Router 1 Command Line Interface:

Router > enable

Router # config t

Enter Configuration commands, one per line. End with CNTL/Z. Router (Config) # interface Fast Ethernet 0/0

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down

Router(config)#int gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

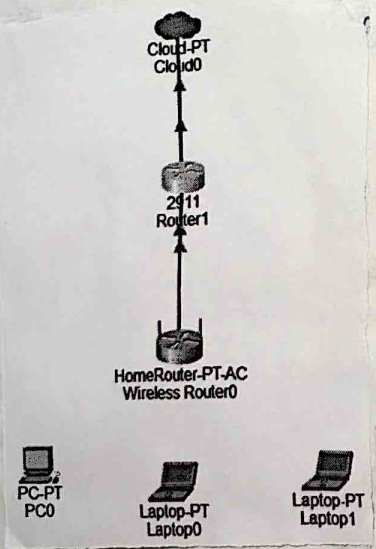
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#copy running-config sta-config
                           ^
% Invalid input detected at '^' marker.

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```




```
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 10.0.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
```

Router (config-if)# ip address 192.168.10.1 255.255.255.0 Router (config-if)# no Shatdown Router (config-if)#

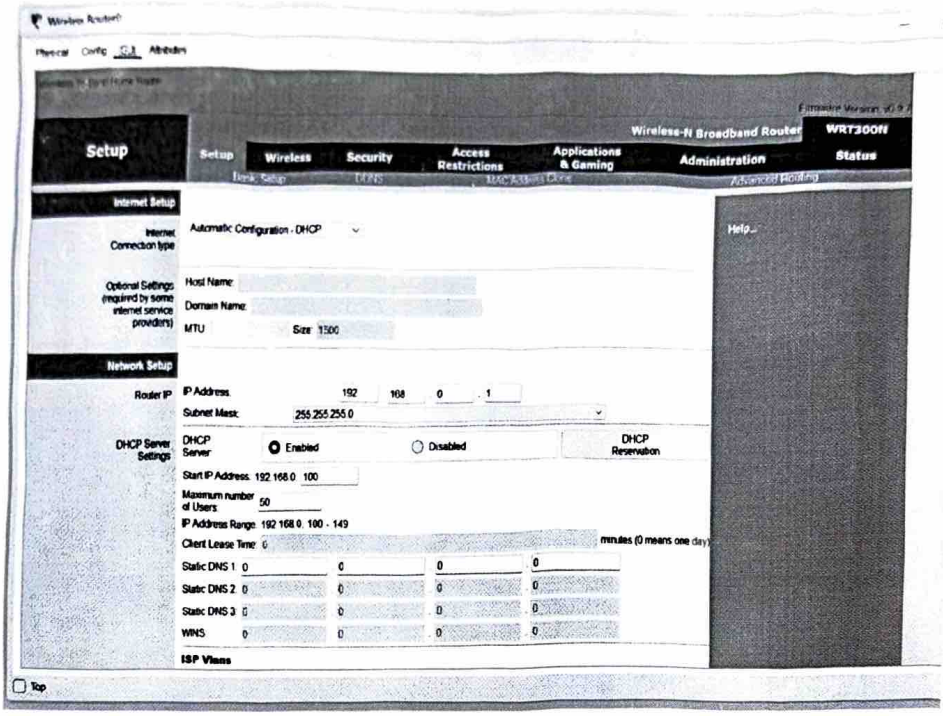
Router (config-if)# ip address 192.168.20.1 255.255.255.0 Router (config-if)# no Shutdown

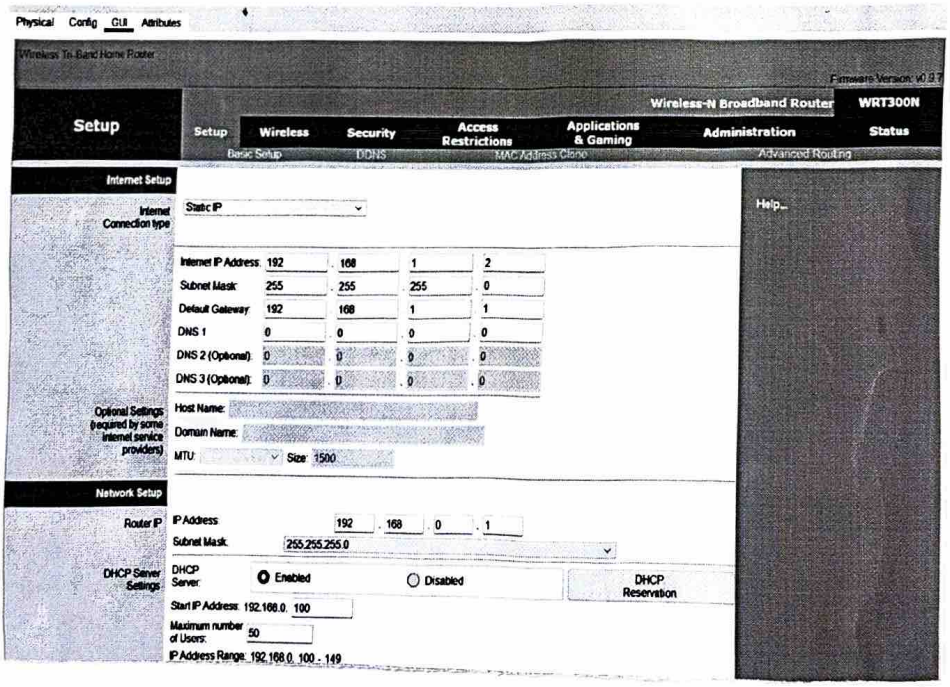
Step-2 (configuring PCs):

1. Assign IP Addresses to every PC in the network.
2. Select the PC, Go to the desktop and Select IP configuration and assign an IP address, Default gateway, Subnet Mask.
3. Assign the default gateway of PC0 as 192.168.10.1
4. Assign the default gateway of PC1 as 192.168.20.1.

Step-3 (connecting PCs with Router):

1. Connect Fast Ethernet 0 Port of PC0 with Fast Ethernet 0/0 Port of Router 1 using a Copper Straigh-Through Cable
2. Connect FastEthernet 0 Port of PC1 with Fast Ethernet 0/1 Port of Router 1 using a Copper Straight-Through Cable.


Physical Config GUI Attributes
Wireless-N Broadband Router WRT300N
Setup
Setup Wireless Security Access Restrictions Applications & Gaming Administration Status
Basic Setup DDNS MAC Address Clone Advanced Routing
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU
Size 1500
Network Setup
Router IP
IP Address: 192 168 0 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server: Enabled Disabled
DHCP Reservation
Start IP Address: 192.168.0. 100
Maximum number of Users: 50
IP Address Range: 192.168.0. 100 - 149
Client Lease Time: 0 minutes (0 means one day)
Static DNS 1: 0 0 0 0
Static DNS 2: 0 0 0 0
Static DNS 3: 0 0 0 0
WINS: 0 0 0 0
ISP Vlans
Help...
Top

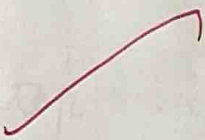

Physical Config GUI Attributes
Wireless-N Broadband Router WRT300N
Setup
Setup Wireless Security Access Restrictions Applications & Gaming Administration Status
Basic Setup DDNS MAC Address Clone Advanced Routing
Internet Setup
Internet Connection type
Static IP
Internet IP Address: 192 168 1 2
Subnet Mask: 255 255 255 0
Default Gateway: 192 168 1 1
DNS 1: 0 0 0 0
DNS 2 (Optional): 0 0 0 0
DNS 3 (Optional): 0 0 0 0
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU
Size 1500
Network Setup
Router IP
IP Address: 192 168 0 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server: Enabled Disabled
DHCP Reservation
Start IP Address: 192.168.0. 100
Maximum number of Users: 50
IP Address Range: 192.168.0. 100 - 149
Help...

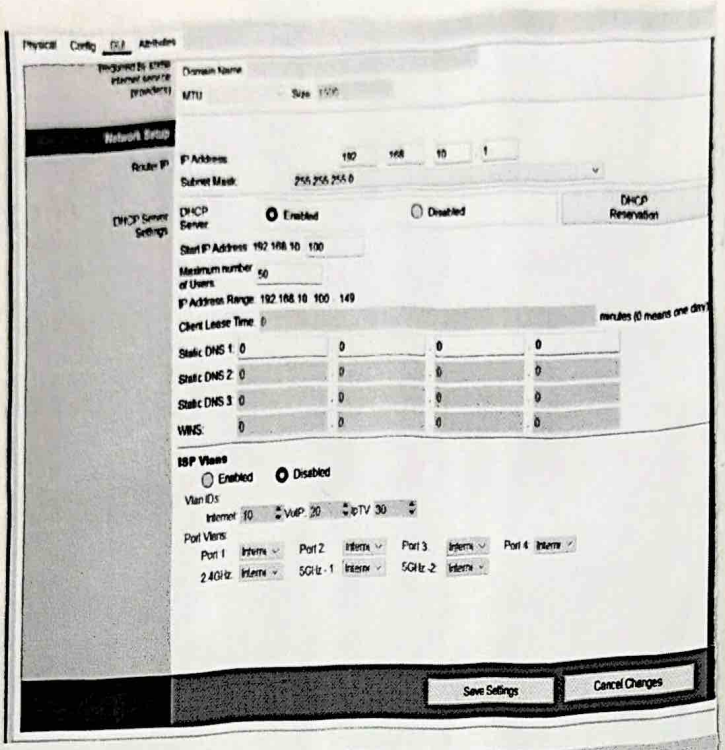
Router configuration Table:

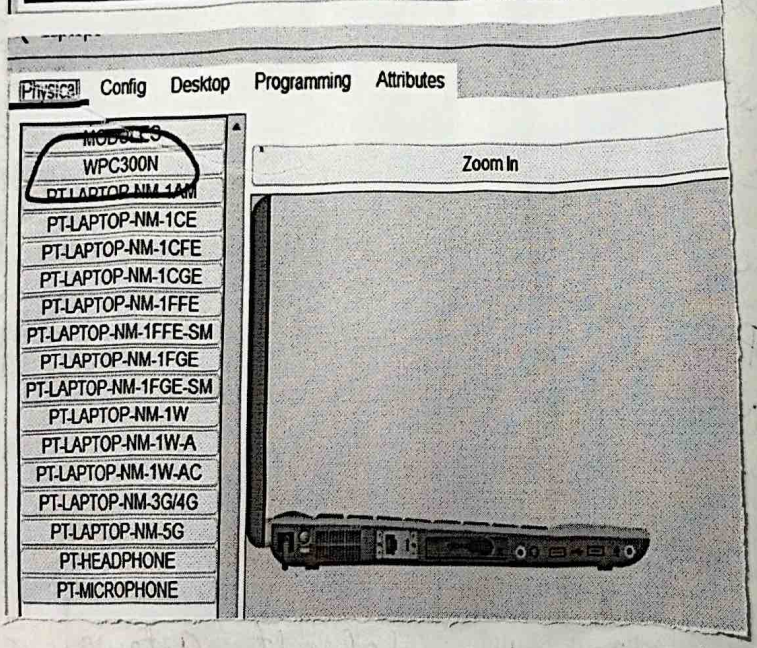
Device Name	IP address Fast Ethernet 0/0	Subnet mask	IP address Fast Ethernet 0/1	Subnet mask
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

Pc configuration Table:

Device Name	IP address	Subnet mask	Gateway
Pc0	192.168.10.2	255.255.255.0	192.168.10.1
Pc1	192.168.20.2	255.255.255.0	192.168.20.1




Physical
Config
GUI
Attributes
Domain Name
MTU
Size
Network Setup
Router IP
IP Address
192 168 10 1
Subnet Mask
255.255.255.0
DHCP Server Settings
DHCP Server
Enabled
Disabled
DHCP Reservation
Start IP Address 192.168.10 100
Maximum number of Users 50
IP Address Range 192.168.10 100 149
Client Lease Time 0
minutes (0 means one day)
Static DNS 1 0 0 0 0
Static DNS 2 0 0 0 0
Static DNS 3 0 0 0 0
WINS 0 0 0 0
ISP Vlans
Enabled
Disabled
Vlan IDs
Internet 10 VoIP 20 IpTV 30
Port Vlans
Port 1 Interne
Port 2 Interne
Port 3 Interne
Port 4 Interne
2.4GHz Interne
5GHz - 1 Interne
5GHz -2 Interne
Save Settings
Cancel Changes


Physical
Config
Desktop
Programming
Attributes
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FFE-SM
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1FGE-SM
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-1W-AC
PT-LAPTOP-NM-3G/4G
PT-LAPTOP-NM-5G
PT-HEADPHONE
PT-MICROPHONE
Zoom In

AIM: b) Design and configure an internetwork using wireless router, DHCP Server and internet cloud.

Addressing Table;

Device	Interface	IP Address	Subnet mask	Default Gateway
PC	Ethernet 0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	Internet	DHCP		
CISCO.Com Server	Ethernet 0	208.67.220.220	255.255.255.0	
Laptop	Wireless 0	DHCP		

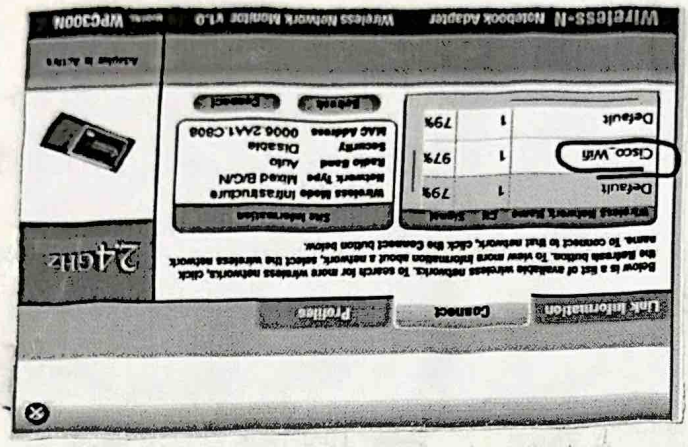
Step 1: Launch Packet Tracer.

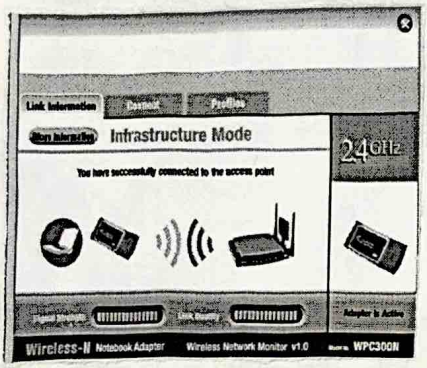
Step 2: Build the topology

a. Add network devices to the workspace.

Using the device selection box, add the network devices to the workspace as shown in the topology diagram.

To place a device onto the workspace, first choose a device type





Font Paragraph

Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>


C:\>ipconfig

Bluetooth Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

Wireless0 Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::260:3EFF:FE29:4546
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.10.101
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                                 192.168.10.1
```

from the Device-Type selection box. Then, click on the desired device model from the Device-Specific selection box. Then, click on the desired device model from the Device-Specific selection box. Finally, click on a location in the workspace to put your device in that location. If you want to cancel your selection, click the Cancel icon for that device. Alternatively, you can click and drag a device from the Device-Specific Selection box onto the workspace.

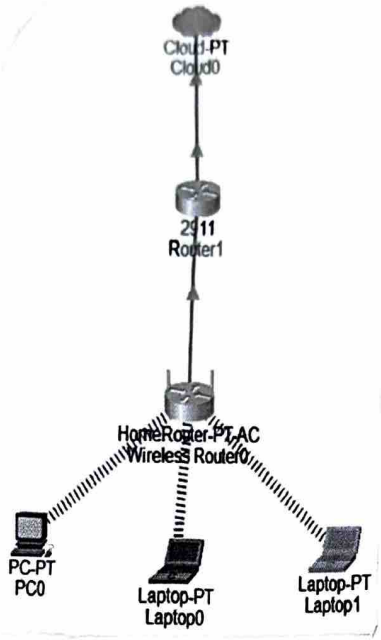
b) Change display names of the network devices.

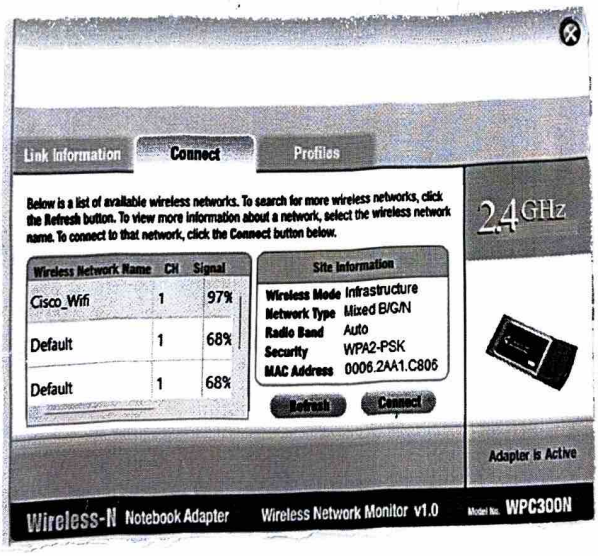
Network devices click on the device icon on the Packet Tracer Logical workspace, click on the config tab in the device configuration window Type the new name of the device.

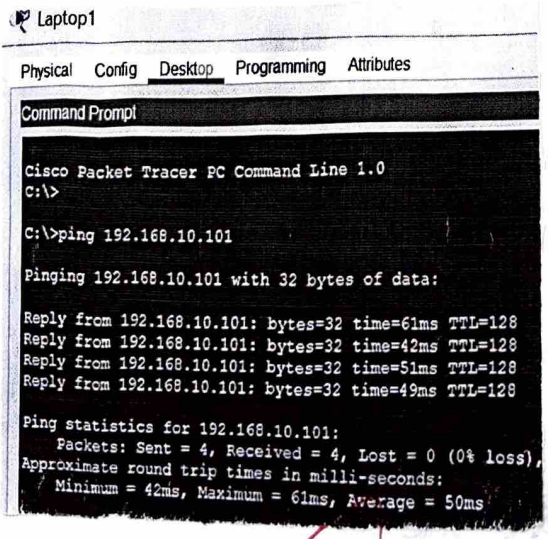
C. Add the Physical cabling between devices on the workspace.

The PC will need a copper straight through Cable to connect to the wireless router.

The wireless router will need a


Cloud-PT
Cloud0
2911
Router1
HomeRouter-PT-AC
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1


Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
Cisco_Wifi
1
97%
Default
1
68%
Default
1
68%
Site Information
Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security WPA2-PSK
MAC Address 0006.2AA1.C806
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
Model No. WPC300N


Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 192.168.10.101
Pinging 192.168.10.101 with 32 bytes of data:
Reply from 192.168.10.101: bytes=32 time=61ms TTL=128
Reply from 192.168.10.101: bytes=32 time=42ms TTL=128
Reply from 192.168.10.101: bytes=32 time=51ms TTL=128
Reply from 192.168.10.101: bytes=32 time=49ms TTL=128
Ping statistics for 192.168.10.101:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 42ms, Maximum = 61ms, Average = 50ms

Copper straight-through cable to connect to the cable modem.

The cable modem will need a coaxial cable to connect to the Internet cloud.

The internet cloud will need copper straight-through cable to connect to the Cisco.com Server.

Part 2: Configure the Network Devices

Step 1: Configure the wireless router

a. Create the wireless network on the wireless router.

Click on the wireless Router icon on the Packet Tracer Logical workspace to open the device configuration window.

In the wireless router configuration window, click on the GUI tab to view configuration options for the wireless router.

Next, click on the wireless tab in the GUI to view the wireless settings. The only setting that needs to be changed from the defaults is the Network Name (SSID)

In the DHCP Server Settings verify that the Enable button is selected and configure the static IP address of the DNS server as 208.67.220.220. as shown in the figure.

b. Click on the Save Settings tab.

Steps 2: Configure the laptop

a. Configure the Laptop to access the wireless network.

Click on the Laptop icon on the Packet Tracer Logical workspace and in the laptop configuration windows select the Physical tap.

In the Physical tap you will need to remove the Ethernet copper module and replace it with the wireless WPC300N module.

Click on the Desktop tab at the top of the Laptop configuration window and select the PC wireless icon

Once the wireless-N Notebook Adapter setting are visible, select the Connect tab. The wireless network "Home Network" should be visible

in the list of wireless networks. Select the network, and click on the connect tab found below the site information pane.

Step 3: Configure the PC

a. Configure the PC for the wired network.

Click on the PC icon on the Packet Tracer Logical workspace and select the Desktop tab and then the IP configuration icon

In the IP configuration window, Select the DCHP radio button as shown in the figure so that the PC will use DCHP to receive an IPV4 address from the wireless router.

Verify that the PC has received an IPV4 address by issuing the ipconfig /all Command from the Command Prompt. The PC should receive an IPV4 address in the 192.168.0.x range

Step: A Configure the Internet cloud

a. Install network modules if necessary

Click on the internet cloud icon on the Packet Tracer Logical workspace and then click on the Physical tab. The cloud device will need two modules if they are not already installed. The PT-CLOUD-NM-1CX which is for the Cable modem Service connection and the PT-CLOUD-NM-1CFE which is for a COPPER Ethernet Cable Connection.

b. Identify the from and to Ports.

Click on the Config tab in the cloud device window. In the left Pane click on cable under CONNECTIONS. In the First drop down box choose Coaxial and in the second drop down box choose Ethernet then Click the Add button to add those as the from Port and TO Port as shown in the figure.

c. Identify the type of Provider

while still in the config tab click Ethernet under INTERFACE in the left Pane.

Steps : 5 Configure the cisco.com Server.

a. Configure the cisco.com Server as a DHCP Server

Click on the cisco.com Server icon on the Packet Tracer Logical workspace and Select the Services tab. Select DHCP from the ~~See~~ SERVICES list in the left.

In the DHCP Configuration window.

- Click On to turn the DCHP Service on
- Pool name : DHCP Pool
- Default Gateway : 208.67.220.220
- DNS Server : 208.67.220.220
- Starting IP Address : 208.67.220.1
- Subnet Mask 255.255.255.0
- Maximum number of users : 50

b) Configure the cisco.com Server as a DNS Server to provide domain name to IPV4 address resolution.

Configure the DNS Service using the following Settings as shown in the figure.

- Click On to Turn the DNS service on
- Name : A Record
- Address : 208.67.220.220

C. Configure the Cisco.com Server Global

Configure the Global Settings of the server as follows:

- Select Static
- Gateway : 208.67.220.1
- DNS Server : 208.67.220.220

d. Configure the Cisco.com Server Fast Ethernet 0 Interface Settings

Configure the Fast Ethernet Interface Settings of the server as follows:

- Select static under IP configuration
- IP Address : 208.67.220.220.
- Subnet mask : 255.255.255.0

Part 3 : Verify Connectivity

Step 1 : Refresh the IPV4 Settings on the PC.

a) Verify that the PC is receiving

IPV4 configuration information from DHCP.

Click on the Pc on the Packet Tracer Logical workspace and then the select the desktop tap of the Pc Configuration window.

Click on the command Prompt icon

In the command Prompt refresh the IP settings by issuing the Commands ipconfig/release and then ipconfig/renew.

b) Test connectivity to the cisco.com Server from the Pc

From the command prompt, issue the command Ping cisco.com,↵

Result :-

To construct a internetworking with cisco routers and switches in Packet Tracer simulator and also wifi, DHCP along with ethernet and wireless connectivity.

Result :- Thus internetworking with routers in CISCO Packet Tracer simulator